T.B.C. : CDS-MATH-24-A Serial No. **589201**	**TEST BOOKLET** **[A]** SERIES	**Time Allowed : Two Hours** **Maximum Marks : 100**

UNION PUBLIC SERVICE COMMISSION

COMBINED DEFENCE SERVICES (CDS) EXAMINATION

TEST BOOKLET

ELEMENTARY MATHEMATICS

ENTER ROLL NO. :

INSTRUCTIONS TO CANDIDATES

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET.

2. Please note that it is the candidate's responsibility to encode and fill in the **Roll Number** and **Test Booklet Series Code A, B, C or D** carefully and without any omission or discrepancy at the appropriate places in the OMR Answer Sheet. Any omission/discrepancy will render the Answer Sheet liable for rejection.

3. You have to enter your Roll Number on the Test Booklet in the Box provided above. **DO NOT write anything else on the Test Booklet.**

4. This Test Booklet contains **100 items (questions)**. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose **ONLY ONE** response for each item.

5. You have to mark all your responses **ONLY on the separate Answer Sheet provided**. See directions in the Answer Sheet.

6. All items carry equal marks.

7. Before you proceed to mark in the Answer Sheet the response to various items in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per instructions sent to you with your Admission Certificate.

8. After you have completed the examination, you should hand over to the Invigilator only the Answer Sheet. You are permitted to take away with you the Test Booklet.

9. Sheets for rough work are appended in the Test Booklet at the end.

10. PENALTY FOR WRONG ANSWERS :
THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.
(i) There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, **one-third (0.33 mark)** of the marks assigned to that question will be deducted as penalty.
(ii) If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
(iii) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE TOLD TO DO SO

SECTION : NUMBER SYSTEM & DIVISIBILITY

1. What is the remainder when 2n is divided by 4 ? What is the last digit in the expansion of [CDS (I) Paper 1 Q3]

(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

2. If log, 6 m and log, 3 log, (r/2) equal to ? = n, then what is Consider the following statements : A number is divisible by 9 if a, + a2 + + + is divisible by [CDS (I) Paper 1 Q4]

(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

3. a, — + a, —a, + as is divisible by [CDS (I) Paper 1 Q9]

(a) I only (b) 2 only (c) Both 1 and 2 (d) Neither I nor 2 What is x

4. What is the value Of 13th observation ? [CDS (I) Paper 1 Q39]

(a) 20 (b) 23 (c) 32 (d) 40 26

5. Consider those numbers between 300 and 400 such that when each number is divi&d by 6, 9 and 12. it leaves 4 as remainder in each case. What is the sum of the numbers ? [CDS (I) Paper 3 Q4]

(a) 692 (b) 764 (c) 1080 (d) 1092 - A 2

6. Which one of the following is a corect statement ? 13, What is the least integral value of k for which the equation has both the roots positive ? [CDS (I) Paper 3 Q12]

(a) 3 years (b) 4 years
(c) 5 years (d) 6 years 8. What is the median of the ages of minor children ?

7. If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [CDS (I) Paper 3 Q10]

(a) 3 years (b) 4 years
(c) 5 years (d) 6 years 8. What is the median of the ages of minor children ?

8. What is the least integral value of k for which the equation has both the roots positive ? [CDS (I) Paper 3 Q13]

(a) 3 years (b) 4 years
(c) 5 years (d) 6 years 8. What is the median of the ages of minor children ?

9. If the highest common factor of two positive integers is 24 then their least Common multiple cannot [CDS (I) Paper 3 Q17]

(a) 72 (b) 216 (c) 372 (d) 600

10. For any integer n, what is the HCF Of integers m = + I and k 9n +4 ? [CDS (I) Paper 3 Q36]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

11. Which one of the following is an infinite set ? [CDS (I) Paper 3 Q38]

(a) {x: x is a whole number less than or equal to 1000} (b) {x: x is a natural number less than 1000}
(c) {x: x is a positive integer less than or equal to 1000} (d) {x: x is an integer and less than 1000}

12. A ladder of 17 ft. length reaches a window Which is 15 ft. above the ground on one side of the street. Keeping its foot at the same point the ladder is turned to the other side of the street and now it reaches a window 8 ft. high. What is the Width of the street ? [CDS (I) Paper 3 Q51]

(a) 23 ft (b) 15 ft. (c) 25 ft. (d) 30ft.

— SPACE FOR ROUGH WORK —

13. What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? (d) 4-5 [CDS (I) Paper 3 Q73]

(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ?

(b) 9th

(c) 15m

(d) I •5m

14. A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (I) Paper 3 Q78]

(a) 100 (b) 90 (c) 80 (d) 75

15. If k is any even positive integer, then (k 2 + 2k) is [CDS (I) Paper 4 Q24]

(a) divisible by 24

(b) d ivisible 8 but may no t be divisible 24

(c) divisible by 4 but may not be divisible by 8

(d) divisible by 2 but may not be divisible by 4

SECTION : HCF, LCM & LOGARITHMS

16. What •is the HCF 4x3 + 3x2y—9xy2 uy3 [CDS (I) Paper 4 Q7]

(a) x—2y of and third side in cm?

(b) g, 2v'fi)

(c) (442

(d) (44ä, 21,110)

17. Rs. 240 Rs. 480 Rs. 800 are in the ratio of 12 : 23. second number ? [CDS (I) Paper 6 Q4]

(a) 44

(b) 55

(c) 66

(d) 77 What is —l? the HCF x3 _ of the 3x2 + (d) None of the above -7Y+10 3x What is the polynomials 1 and Ilx +4y —10 9

18. Which one of the following is a prime number ? [CDS (I) Paper 6 Q90]

(a) 161

(b) 171

(c) 173

(d) 221 What is (a) 16 equal to ? What is log,o — + log,o ¯ + log.o — + up to 8 terms equal to ? (a) (c) (d)

19. equal to ? [CDS (I) Paper 7 Q41]

(a) 5/4

(b) 6/5

(c) 25/31

(d) 31/42 The sum of two numbers is 232 and their HCF is

20. If 18 is added to the number, the digits are reversed. The number is equal to [CDS (I) Paper 8 Q8]

(a) 26

(b) 35

(c) 53

(d) 62 If log, o a = p and loglo b = q, then what is loglo (aP bq) equal to ? (a) P2 + q2 (c) p2q2 q

21. If (x+l) is the HCF of Ax2+Bx+C and Bx2 + Ax + C where A B, then the value of C is [CDS (I) Paper 14 Q14]

(a) I only

(b) 2 only

(c) I and 2 only l, 2 and Y is/are 14. If

(d) 2 minutes

SECTION : PERCENTAGES, PROFIT, LOSS & DISCOUNT

— SPACE FOR ROUGH WORK —

22. If the height of a cone•is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? [CDS (I) Paper 4 Q71]

(a) 100/3 (b) 40 (c) 50 (d) 200/3

23. If 28 [CDS (I) Paper 4 Q93]

(a) The area Of the circle is e qual to that of square
(b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square
(d) No conclusiion can be drawn 87• If

24. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) Paper 4 Q92]

(a) The area Of the circle is e qual to that of square
(b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square
(d) No conclusiion can be drawn 87• If

25. D 3k E BC 3 ADE 3 cm2 I EC ? [CDS (I) Paper 5 Q92]

(a) 16 cm
(b) 20 em
(c) 25 cm
(d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

26. 16Jä -8T 16Jä - 27 -A) [CDS (I) Paper 5 Q95]

(a) 16 cm
(b) 20 em
(c) 25 cm
(d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ? (a) 1-5 C m (d) 2-1 cm qx 9,680 10,648 I (c) 15%

27. Two students A and B solve an equation Of the form ± px + q = O. A starts with a %T0ng value of p and obtains the roots as 2 and [CDS (I) Paper 7 Q63]

(a) p (b) 3 and —4 (c) —3 and—4 (d) —3 and 4

28. What is 5% of of 500 ? [CDS (I) Paper 8 Q17]

(a) 12.5 (b) 25 (c) 1.25 (d) 6.25

29. A man buys a television set which lists for Rs. 5,000 at 10% discount. He gets an additional 2% discount (after the first discount) for paying cash. What does he actually pay for the set ? [CDS (I) Paper 8 Q19]

(a) Rs. 4,410 (b) Rs. 4,400 (c) Rs. 4,000 (d) Rs. 4,500

30. -Ai if one Of the Of the — bx + c O the square at the then Which one the following is ? 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf — 5x2 — 81 ia ta) equgi to ? Let A : % is odd integer) and B — : xa — 8x + % — [CDS (I) Paper 9 Q22]

(a) 5% (b) 8% (c) 10% (d) 12.5%

31. 1%en which one of the ig ? (b) (41 ACB sum Of numbers 7 and sum Of 20. their ' is [CDS (I) Paper 9 Q1]

(a) Rs. 1,200 (b) Rs. 1,440 (c) Rs. 1,500 (d) Rs. 1,800

SECTION : SIMPLE & COMPOUND INTEREST, RATIO & MIXTURES

32. The wages, of labourers in a factory has increased in the ratio 22 : 25 and their number decreased in the ratio 3 : [CDS (I) Paper 3 Q34]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

— SPACE FOR ROUGH WORK —

33. If two corresponding sides Of two similar triangles, are in the ratio 9 : 4, then What • the ratio Of their areas ? (c) 81 16 / 26A (d' 27 : 8 28 [CDS (I) Paper 4 Q89]

(a) The area Of the circle is e qual to that of square
(b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square
(d) No conclusiion can be drawn 87• If

34. The ratio of two complementary angles is I : [CDS (I) Paper 7 Q91]

(a) 64 minutes
(b) 80 minutes
(c) 96 minutes
(d) 120 minutes The pair of numbers which are relatively prime to each other is :

35. The ratio of two complementary angles is I : [CDS (I) Paper 7 Q90]

(a) 64 minutes
(b) 80 minutes
(c) 96 minutes
(d) 120 minutes The pair of numbers which are relatively prime to each other is :

36. Sex ratio is defined as the number of females per 1000 males. In a place the total inhabitants are out of which 935,000 are females. What is the sex ratio for the place ? [CDS (I) Paper 8 Q20]

(a) 935 (b) 1000 (c) 1935 (d) 9350 6 2 23 2

37. The speeds of A, B and C are in the ratio 50 : 45 : [CDS (I) Paper 14 Q2]

(a) 43-2%
(b) 56-8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ?
(c) 5<k$11 21. The number of pairs
(d) Zero

38. A water tank, open at the top, is hemi- spherical at the bottom and cylindrical above it. Ihe radius is 12m and the capacity is 3312rt m3. The ratio of the surface areas of the spherical and cylindrical portions is - A [CDS (I) Paper 14 Q73]

(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

39. The area of a trapezium is 336cm2. If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is [CDS (I) Paper 14 Q78]

(a) 20 cm (b) 22 cm (c) 24 cm (d) 26 cm

SECTION : TIME, SPEED & DISTANCE, TIME & WORK

40. If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the sarru in 2 days, what is the time taken by 15 men and 20 boys in doing the same type of work ? [CDS (I) Paper 3 Q32]

(a) 4 days (b) 5 days (c) 6 days (d) 7 days

41. The inner and outer radii of a 7 m long hollow iron right circular cylindrical pipe are 2 cm and 4 cm respectively. If 1000 cni3 of iron weighs 5 kg, what is the weight of the pipe ? [CDS (I) Paper 3 Q86]

(a) 264 kg (b) 132 kg (c) 396 kg (d) None of the above

— SPACE FOR ROUGH WORK —

42. What is, the mean proportional between and (27 ? (b) 146 A stock Of food is enough for 240 men for 48 days. How long will the same stock last for 160 men ? [CDS (I) Paper 7 Q33]

(a) 54 days

(b) 60 days

(c) 64 days

(d) 72 days 9 2 3 = 2 and If x+Y x+y x —Y then what is ¯ equal to ? (a) 3/2 (c) 2/3 (d) 1/5 4

43. A car is traveling at a constant rate of 45 km per hour. The distance travelled by car from 10.40 am to 1.00 pm is [CDS (I) Paper 8 Q23]

(a) 165 km

(b) 150 km

(c) 120 km

(d) 105 km A number consists of two digits whose sum is

44. Three planets revolve round the sun once in 200, 250, days respectively in their own orbits. When do they all come relatively to the same position as at a certain point of time in their orbits ? [CDS (I) Paper 8 Q28]

(a) After 3000 days (b) After 2000 days (c) After 1500 days (d) After 1200 days 8

45. men and 1 woman can complete a piece of work in 14 days, while 4 women and 2 men can do the same work in 8 days. If a man gets 90 per day, what should be the wages per day Of a woman? (d) 48 72 e 135 [CDS (I) Paper 10 Q30]

(a) HCF

(b) HCF

(c) HCF

(d) None Of the above If b is the largest square divisor Of c and a2 divides C, then which one of the following is correct?

46. A and B are two taps which can fill a tank individually in 10 minutes and 20 minutes respectively. However, there is .a leakage at the bottom, which can empty a filled tank in 40 minutes. If the tank is empty initially, how much time •c will both the taps take to fill the tank with leakage ? [CDS (I) Paper 14 Q17]

(a) 2 minutes (b) mihutes (c) 5 minutes (d) 8 minutes

47. By increasing the speed of his car by 15 km/hour, a person covers 300 km distance by taking an hour less than before. The original speed of the car [CDS (I) Paper 14 Q28]

(a) 45 km/hour (b) 50 km/hour (c) 60 kni/hour (d) 75 km/hour

48. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third mysition is [CDS (I) Paper 14 Q11]

(a) 1000m (b) 800m (c) 750m (d) 600 m

49. An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from some point on the ground are 600 and 450 respectively. Then the vertical distance between the two planes is [CDS (I) Paper 14 Q43]

(a) 100066-1

(b) 10000m

(c) 100+-6

(d) 24.16 m 45. If tano + = 2, then tang is equal to 3 4 5 4 2 5

SECTION : ALGEBRA, POLYNOMIALS & LINEAR EQUATIONS

50. From a rectangular metal sheet Of sides 25 cm and 20 em, a circular sheet as large as possible is cut-off. What is the area of the remaining sheet ? [CDS (I) Paper 1 Q49]

(a) 186 (b) 144 (c) 93 cm2 (d) 72 cm2

— SPACE FOR ROUGH WORK —

51. (d) x —y = 10 [CDS (I) Paper 1 Q76]

(a) x —y = 10 78. 79. 80. 81. 82. 83. 84. sin x cos x 1/2 t, sin x — cos x DE t

(b) *rut A log IO

(c) Tt/2

(d) Tt/3

52. Assertion [CDS (I) Paper 1 Q7]

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true 98. Assertion

53. What was the original wage bill of the factory if the present bill is Rs. 5,000/- ? (b) Rs. 6000/- (c) Rs. 8,000,'- (d) None of the above [CDS (I) Paper 3 Q2]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

54. If a quantity y varies as the sum of three quantities of which the first varies as x, the second varies as —x+x2, the third varies as x3 —p then what is y equal to ? [CDS (I) Paper 3 Q35]

(a) where k is a constant (b) kx+lx2+nu3, where k, l, m are constants

(c) kP where k is a constant (d) kx where k is a constant - A 10

55. What is the „ + sin275ø ? + sin2250 + [CDS (I) Paper 3 Q41]

(a) tan2150 + + tan2250 tan2750 (b) cos2150 + cos220e + cos2250 . cos2750

(c) + cot225• — + cot2750 (d) sec2200 + sec2250

56. A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 [CDS (I) Paper 3 Q76]

(a) cm3 (b) 3006 cm3 (c) cm3 1078 (d) cm3 - A 22

57. Smaller lead shots are to be prepared by using the material of a spherical lead shot of radius I cm. Some possibilities are listed in the statements given below: . I. The material is just sufficient to prepare 8 shots each of radius cm. [CDS (I) Paper 3 Q77]

(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ?

(b) 9th

(c) 15m

(d) I •5m

58. A shot of radius 0-75 cm and a second shot of radius 0-8 cm can be prepared from the available materiaL Which of the above statements is/ are correct ? [CDS (I) Paper 3 Q2]

(a) 1 Only (b) 2 Only (c) Both I and 2 (d) Neither nor 2

59. The HCFof x —y and xlO—y10 is Which of the statements given above Correct ? are [CDS (I) Paper 3 Q4]

(a) 1 and 2 (b) 2 and 3 (c) 1 and 4 (d) 2 and 4 26

60. What is the sum of the squarés of all the six numbers ? [CDS (I) Paper 3 Q98]

(a) 40 (b) 42 (c) 45 (d) 48

61. What are the factors Of [CDS (I) Paper 4 Q4]

(a) (x —2y—4) and (x —2Y+2) (b) (x —y and (x —4 y + 4) (c) (x —y and (x —4 y —4) (d) (x (x +2Y+2) 26k /

— SPACE FOR ROUGH WORK —

SECTION : QUADRATIC EQUATIONS & PROGRESSIONS

62. B starts with a wrong value Of q and gets the roots as 2 —9. What the correct roots of the equation ? [CDS (I) Paper 7 Q6]

(a) 3 and — 4 (b) —3 and 4 (c) —3 and 4 (d) 3 and 4 64. of

63. What is the value Of 1001xlOOlxiOOl+0999x0999xo•999 1001x x0999+0999xo-999 If p(x) is a common multiple of degree 6 Of the (x) = x' + x2 — x — I and g(x) — — x2 + X— I then which Of the following is correct? [CDS (I) Paper 7 Q62]

(a) p (b) 3 and —4 (c) —3 and—4 (d) —3 and 4

64. The roots of the equation are always unequal and real, Which Of the above statements is/ are correct? [CDS (I) Paper 10 Q2]

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2 and ß are the roots or the

65. If the sum of the roots of ax + bx+c—O is equal to the sum of the squares of their reciprocals, then which one Of the following relations is correct? [CDS (I) Paper 15 Q16]

(a) bc 2 = 2a2c (b) ac2+ 2b 2 a (c) ab2+ bc 2' a 2c (d) a2+b2+c2=I

66. If p and q are the roots ofx2+px+q=0, then which one of the following is correct ? (b) p -loniy = —2 or O = —2 Ohly [CDS (I) Paper 16 Q20]

(a) p -loniy 20. 6 If p and q are the roots of x2+px+q = 0, then which one of the following is correct ? remaining work ?

(b) 6 hours

(c) 12 hours

(d) hours

SECTION : TRIGONOMETRY & HEIGHTS AND DISTANCES

67. If sin x cos x = 1/2, then what is the value of sin x — cos x ? nt lat of Iftan2 y cosec2 x — 1 tana y, then which 81. one of the following is correct ? (b) x = 2y (c) y — 2x [CDS (I) Paper 1 Q80]

(a) The angle bisector of angle A (b) The angle trisector Of angle A

(c) bisector of angle A (d) None of the above What is log

68. 22 COS x cos x 2 , which one I + cosec x cosecx — of the following is one Of the values [CDS (I) Paper 1 Q83]

(a) The angle bisector of angle A (b) The angle trisector Of angle A

(c) bisector of angle A (d) None of the above What is log

69. sin x cos x 1/2 t, sin x — cos x 80. tan2 y coseca x — 1 = tan2 y t, 81. (d) x —y = 10 [CDS (I) Paper 1 Q84]

(a) x —y = 10 78. 79. 80. 81. 82. 83. 84. sin x cos x 1/2 t, sin x — cos x DE t

(b) *rut A log IO

(c) Tt/2

(d) Tt/3

70. 150m 300 I 450 [CDS (I) Paper 1 Q89]

(a) 59•4 m

(b) 61 •4m

(c) 62•4 m

(d) 63•4 m 90. w, (c) 2vif 24 91. sin x + sin y a cos x + cos y = b t, sinx siny+cosx (a) (b) (c) 25 a + b — ab

— SPACE FOR ROUGH WORK —

71. 150m gr 300 I 450 [CDS (I) Paper 1 Q91]

(a) 59•4 m

(b) 61 •4m

(c) 62•4 m

(d) 63•4 m (a) (b) (c) (d) 24 sin x + sin y — a cos x + Cosy = b t, sinx siny+cosx (a) (b) (c) (d) a + b — ab 2

72. Which one of the following is correct ? [CDS (I) Paper 3 Q11]

(a) There is only one g with 00 < < 900 such that sin = a, where a is a real number

(b) There is more than one O with 00 < e 900 such that sin = a, where a is a real number - A 4

(c) There is no 9 with 00 < 900 such that sin a. where a is a real

(d) There are exactly two Os with < 900 such that sin = a, where a is a real number

73. If sinO+cosO=i, then what is the value of sine dose? 2 (d) 1/2 I + sin a [CDS (I) Paper 4 Q46]

(a) 1/2 47. (b) 60 (c) 45 (d) 30

74. What is equal to? I—sin0 (b) secG+ tane (c) cosece + cot é (d) cosec — c Ot [CDS (I) Paper 4 Q47]

(a) 1/2 47. (b) 60 (c) 45 (d) 30

75. The shadow Of a tower is 15 m when the Sun's altituqe is [CDS (I) Paper 4 Q48]

(a) 1/2 47. (b) 60 (c) 45 (d) 30

76. sm t? I — sine [CDS (I) Paper 4 Q49]

(a) 9 — tan 9

(b) sece+ tan 8

(c) cosece + cot 9

(d) cosecO — cot 9 30' t, 15 m 60• t, (d) 100 (b) (c) 8905 83 192 33 88-55 8755 89-55 26A IS (c)

77. 5 mm [CDS (I) Paper 5 Q94]

(a) 22 m

(b) 20 m

(c) 18 m 95. qt«rq (cm2 t ? 16Jä -8T 16Jä - (a) (b) (c)

(d) 16 NG 27 - ' g 96. (a) 3-5 km (b) km (c) 5•5 km (d) 6-5 km cot 19 cot 200 cot 700 cot 75' '

78. sin cos (9-20) 6 0 — cos e — cos I — sin2 B cos2 e 2-5 kmph 3aqå 3-25 kmph t I 4 36 [CDS (I) Paper 5 Q97]

(a) 16 cm (b) 20 em

(c) 25 cm (d) 30 cm ABC !3T3-å AB AC qi D 3k E BC 3 ADE 3 cm2 I EC ?

SECTION : GEOMETRY: LINES, ANGLES, TRIANGLES & POLYGONS

79. Two sides Of a parallelogram are IO cm and IS cm. If the altitude corresponding to the side Of length 15 cm is 5 crn, then what is the altitude to the side Of length [CDS (I) Paper 1 Q48]

(a) 1/64 What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ?

(b) 47

(c) 43

(d) 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ?

— SPACE FOR ROUGH WORK —

80. In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? [CDS (I) Paper 1 Q75]

(a) 500 (b) 600 (c) 700 (d) 800

81. Assertion [CDS (I) Paper 1 Q98]

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true 98. Assertion

82. ABC is a triangle. Let D, E denote the mid points Of BC, CA respectively. Let AD and BE intersect at G Let O be a point on AD such that AO : OD 2 : [CDS (I) Paper 1 Q99]

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true 98. Assertion

83. Staternent—I : I.Rt LMNbea triangle. Let P, Q be the mid- points of the sides LM, LN respectively. If PQ2&MP2+NQ2, then LMN is a fight angled triangle right angled at L. Statement—II : If in a triangle ABC, .4B2 > BC + CA2, then LACB is obtuse. Which of the following is correct in the light of the above statements ? [CDS (I) Paper 3 Q6]

(a) Both statement — I and statement 11 are correct and statement — 11 is the reason for statement — I

(b) Both statement — I and statement — 11 are corect and statemeht — 11 is not the reason for statement — I

(c) Statement—I is true, but state: ment — 11 is false

(d) Statement — I is false, but state— ment — 11 is true 2

84. What is tlE median of the ages Of minor children ? (b) 5 years (c) 7 years (d) Cannot determined due to in- sufficient data [CDS (I) Paper 3 Q8]

(a) 3 years (b) 4 years

(c) 5 years (d) 6 years 8. What is the median of the ages of minor children ?

85. The median BD of ghe triangle ABC meets at D. If then which one Of the following is correct ? [CDS (I) Paper 3 Q20]

(a) ZACB = I right angle (b) LBAC = I right angle (c) LABC I right angle (d) None of ttw abové - A 6

86. ABC is a right triangle with right angle at A. If the value of tan B = any.real k the length of the hypotenuse is of the form (b) 2k (d) 9k (c) 5k 10 [CDS (I) Paper 3 Q39]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

87. The perimeter of a triangle is greater than the sum of the lengths of its three medians. Which of the statemepts given above is/are correct ? [CDS (I) Paper 3 Q2]

(a) only (b) 2 only (c) Both 1 and 2 (d) Neither 1 nor 2 - A 16

88. Consider the follOW'ing statements • If two mangles are equi- angular, then they are similar. [CDS (I) Paper 4 Q88]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

SECTION : GEOMETRY: CIRCLES, CHORDS & TANGENTS

— SPACE FOR ROUGH WORK —

89. What is the of centres of circles which touch a given line at a given point ? [CDS (I) Paper 3 Q2]

(a) A line pen*ndicular to thc given line, passing through the given
(b) A line parallel to the given line
(c) A circle tangent to the given line at the given point
(d) A closed curve other -than a circle

90. If ABC is a triangle right angled at C and having u units, v units, w units as the lengths of its sides opposite to the veltices A, B, C respectively, then what is the tangent of the angle at A + tangent of the angle at B equal to ? (d) [CDS (I) Paper 3 Q37]

(a) 4 days (b) 5 days (c) 6 days (d) days 33. If of

91. 31-5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (I) Paper 3 Q69]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A 20

92. A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? [CDS (I) Paper 4 Q74]

(a) 9 Crn (b) 8 Crn (c) 7 Crn (d) 6 Ctn

SECTION : MENSURATION 2D: AREAS & PERIMETERS

93. ABCD is a square. p. Q, R, S are points on the sidesAB, BC, CD, DA respectively such that CR=DS. What is ZSPQ equal ? [CDS (I) Paper 3 Q49]

(a) 300 (b) 450 (c) 600 (d) 900

94. A parallelogram and a rectangle stand on the same base and on the same side of the base with the same height. If 11, be the perimeters of the parallelogram and the rectangle respectively, then which one of the following is correct ? [CDS (I) Paper 3 Q58]

(a) 11 12 (b) 11 = 12 (c) 12 but 11 *212 (d) 11 -212

95. Which one of the following numbers is not a square of any natural ? [CDS (I) Paper 3 Q88]

(a) 5041 (b) 9852 (c) 1936 (d) 6241 T-DETA.K.FN - A 26

96. What is the maximum area of a rectangle, the IVimeter of which is 18cm? [CDS (I) Paper 3 Q89]

(a) 2025 cm2 (b) 20-00 cm2 (c) 19-75 cm2 (d) 19-60 cm2

SECTION : MENSURATION 3D: SURFACE AREAS & VOLUMES

97. Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? [CDS (I) Paper 1 Q50]

(a) 300 cml (b) 350 cm2 (c) 375 cm2 (d) 400 cm2

98. What is volume of the frustum of a cone with height h and radii r, , ? [CDS (I) Paper 3 Q74]

(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I •5m

99. A cylinder having base of circumference 60 cm is rolling without sliding at a rate of 5 rounds per second. How much dis- tance will the cylinder roll in 5 seconds ? [CDS (I) Paper 3 Q75]

(a) 4-5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I •5m

— SPACE FOR ROUGH WORK —

100. A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(I cm3 of the metal weighs 9 g) ? [CDS (I) Paper 3 Q79]

(a) 54 g (b) 366 g (c) 122 g (d) 1087t g

— SPACE FOR ROUGH WORK —

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